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APUL

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# PHYSICS

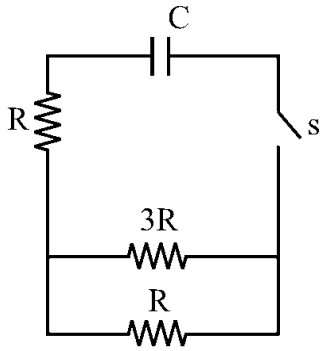
1) A capacitor of capacitance $C_1$ = 1μF can withstand maximum voltage $V_1$ = 6 kV (kilo-volt) and another capacitor of capacitance $C_2$ = 3μF can withstand maximum voltage $V_2$ = 4kV. When the two capacitors are connected in series, the combined system can withstand a maximum voltage of :-

(1) 4 kV
(2) 6 kV
(3) 8 kV
(4) 10 kV

2) A capacitor 10 μF charged to 50V is joined to another uncharged 50μF capacitor. Find the loss in energy.

(1) $1.04 \times 10^{-2}$ J
(2) $4.01 \times 10^{-2}$ J
(3) $6.25 \times 10^{-4}$ J
(4) $1.64 \times 10^{-4}$ J

3) In the figure, capacitor is completely charged and switch is closed at t = 0. The time after which the current from the capacitor becomes $1/4^{th}$ of its maximum value will be :-




(1) 2 RC ln2
(2) $\frac{7RC}{4}$ ln2
(3) RC ln2
(4) $\frac{7RC}{2}$ ln2

4) A 4 μF capacitor, a resistance of 2.5 MΩ is in series with 12 V battery. Find the time after which the potential difference across the capacitor is 3 times the potential difference across the resistor. [Given ℓn(2) = 0.693]

(1) 13.86 s
(2) 6.93 s

(3) 7 s

(4) 14 s

5) A parallel plate capacitor filled with a material of dielectric constant K is charged to a certain voltage. First the battery is removed then the dielectric material is removed. Then which of the following is true :-
(i) Capacitance decrease by factor K
(ii) Electric field reduced by factor K
(iii) Voltage across capacitor increases by factor K
(iv) Charge stored in capacitor increases by factor K

(1) (i) and (ii)

(2) (i) and (iii)

(3) (ii) and (iii)

(4) (ii) and (iv)

6) A capacitor of capacity 4μF is charged to 80V and another capacitor of capacity 6μF is charged to 30V. When they are connected , then the energy lost by 4μF capacitor is

(1) 7.8 mJ

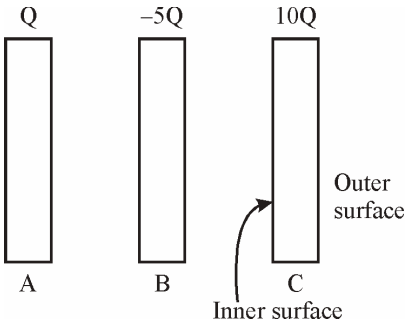
(2) 4.6 mJ

(3) 3.2 mJ

(4) 2.5 mJ

7)

Three very large plates are given charges as shown in the figure. If the cross-sectional area of each plate is the same, the final charge distribution on plate C is :-




(1) +5Q on the inner surface, +5Q on the outer surface

(2) +6Q on the inner surface, +4Q on the outer surface

(3) +7Q on the inner surface, +3Q on the outer surface

(4) +8Q on the inner surface, +2Q on the outer surface

8) A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2ms. The power delivered to the patient is :-

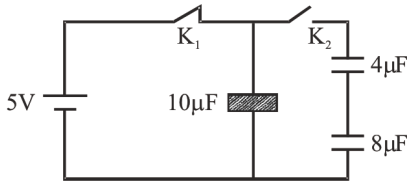
(1) 45 kW

(2) 90 kW

(3) 180 kW

(4) 360 kW

9) In given circuit capacitance of a capacitor with medium (k = 2) is 10μF initially $k_1$ is closed and $k_2$ is open. Now $K_1$ is open and $K_2$ is closed and medium from 10μF is removed and filled in a capacitor of capacitance 4μF then what will be final change of 4μF capacitor :-



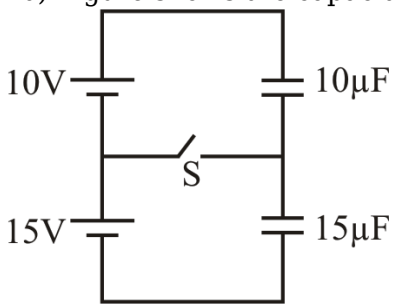

(1) $\frac{200}{9}\mu C$

(2) $\frac{100}{9}\mu C$

(3) $\frac{50}{3}\mu C$

(4) $\frac{200}{3}\mu C$

10) Figure shows the capacitors in steady state, with S open. When S is closed :-




(1) Both batteries supply energy & heat is lost.

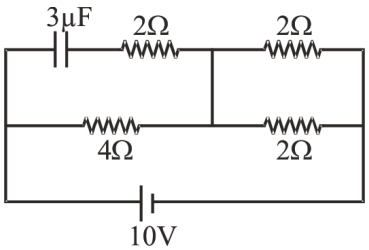
(2) Both batteries supply energy to increase the energy of both the capacitors separately

(3) One battery supplies & other absorbs energy and no heat loss takes place.

(4) One battery supplies energy, other absorbs and heat is lost in the process.

11)

Find charge stored capacitor in steady state condition :-




(1) 24 μC

(2) 12 μC

(3) 6 μC

(4) 4 μC

12) A parallel plate capacitor has plate area A and separation d. It is charged to a potential

difference $V_0$. The charging battery is disconnected and the plates are pulled apart to three times the initial separation. The work required to separated the plates is :-

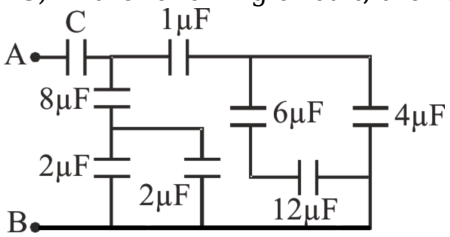
(1) $\frac{3\varepsilon_0 A V_0^2}{d}$

(2) $\frac{\varepsilon_0 A V_0^2}{2d}$

(3) $\frac{\varepsilon_0 A V_0^2}{3d}$

(4) $\frac{\varepsilon_0 A V_0^2}{d}$

13) In the following circuit, the resultant capacitance between A and B is 1 μF. Then value of C is




(1) $\frac{32}{11}\mu F$

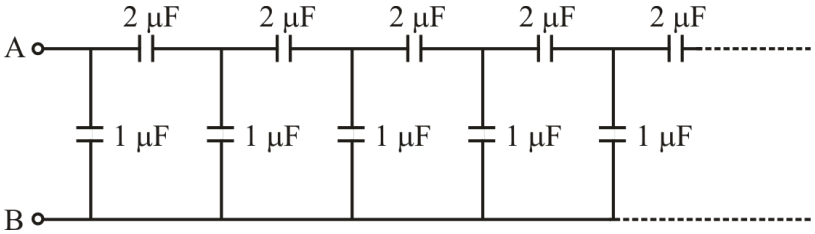
(2) $\frac{11}{32}\mu F$

(3) $\frac{23}{32}\mu F$

(4) $\frac{32}{23}\mu F$

14)

Find the equivalent capacitance of the infinite ladder shown in figure between the points A & B.



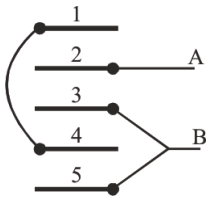

(1) 1μF

(2) 4μF

(3) 2μF

(4) 8μF

15) Five identical metal plates 1, 2, 3, 4 and 5 each of area A on one side are fixed parallel and equidistant (d) to each other. The plates 1 and 4 are joined by a conductor, and plates 3 and 5 are also joined by a conductor as shown in figure. Then, the capacitance of this system between A and B




is-

(1) $\frac{5\varepsilon_0 A}{d}$

(2) $\frac{4\varepsilon_0 A}{d}$

(3) $\frac{5\varepsilon_0 A}{3d}$

(4) None of the above

16) The plates of a capacitor are charged to a potential difference of 100 V and then connected across a resistor. The potential difference across the capacitor decays exponentially with respect to time. After one second. the potential difference between the plates of the capacitor is 80 V. What is the fraction of the stored energy which has been dissipated ?

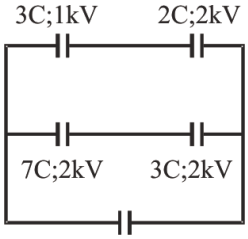
(1) $\frac{1}{25}$

(2) $\frac{18}{25}$

(3) $\frac{9}{25}$

(4) $\frac{27}{25}$

17) The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down ?



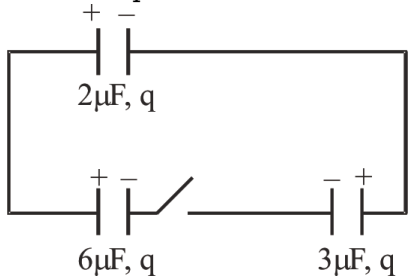

(1) 2.5 kV

(2) 10/3 kV

(3) 3 kV

(4) 1 kV

18) In given circuit all capacitors has same charge q when switch is closed then fined charge flow through switch



(1) $\frac{q}{2}$

(2) $\frac{2q}{3}$

(3) $\frac{q}{3}$

(4) $\frac{q}{4}$

19) A series LCR circuit containg 5.0 H inductor 20μF capacitor and 50Ω resistor is connected to 230 V variable frequency AC source. The angular frequency of source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :

(1) 85 rad/s and 115 rad/s

(2) 75 rad/s and 125 rad/s

(3) 95 rad/s and 105 rad/s

(4) 98 rad/s and 102 rad/s

20) A coil has an inductance of 0.7 henry and is joined in series with a resistance of 220 Ω. When the alternating emf of 220 V at 50 Hz is applied to it then the phase through which current lags behind the applied emf and the wattless component of current in the circuit will be respectively:–

(1) 30°, 1 A

(2) 45°, 0.5 A

(3) 60°, 1.5 A

(4) None of these

21) Match the physical quantities given in column-I with the parameters they depend on as given in column-II.

| | Column-I | | Column-II |
|---|---|---|---|
| (A) | Inductance of a coil | (p) | Depends on resistivity |
| (B) | Capacitance | (q) | Depends on shape |
| (C) | Impedance of a coil | (r) | Depends on medium inserted |
| (D) | Reactance of a capacitor | (s) | Depends on external AC voltage source |

(1) (A)-q,r (B)-q,r (C)-p,q,r,s (D)-q,r,s

(2) (A)-q,r (B)-q,r,s (C)-p,q,r (D)-q,r,s

(3) (A)-q,r,s (B)-q,r (C)-p,q,r (D)-q,r,s

(4) (A)-q,r (B)-q,r (C)-p,q,r (D)-p,q,r,s

22) For current I = 12 + 3 sin ωt + 4 cos ωt
Match column I with column II

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (A) | Peak value of current | (P) | 16 |

| | | | |
|---|---|---|---|
| (B) | Instantaneous value of current at t = 0 | (Q) | 17 |
| (C) | Effective value of current | (R) | 5 |
| (D) | Amplitude of current | (S) | $\sqrt{\frac{313}{2}}$ |
| | | (T) | 12 |

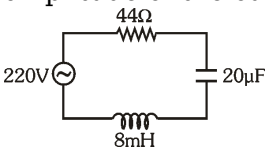
(1) A-R, B-P, C-S, D-Q

(2) A-Q, B-P, C-T, D-R

(3) A-Q, B-P, C-S, D-R

(4) A-R, B-T, C-T, D-R

23) For the series LCR circuit shown in the figure. What is the resonance angular frequency and the amplitude of the current at the resonating frequency :




(1) 2500 rad/sec, $5\sqrt{2}$A

(2) 2500 rad/sec, 5A

(3) 2500 rad/sec, $\frac{5}{\sqrt{2}}$A

(4) 25 rad/sec, $5\sqrt{2}$A

24) The self inductance of a choke coil is 10 mH. When it is connected with a 10V dc source, then the loss of power is 20 watt. When it is connected with 10 volt ac source, then loss of power is 10 watt. The frequency of ac source will be approx

(1) 50 Hz

(2) 60 Hz

(3) 80 Hz

(4) 100 Hz

25) A box P and a coil Q are connected in series with an ac source of variable frequency. The emf of source is 10 V. Box P contains a capacitance of 1μF in series with a resistance of 32Ω while coil Q has a self-inductance 4.9 mH and resistance 68Ω . The frequency is adjusted so that the maximum current flows in P and Q. The impedance of P and Q at this frequency will be :-

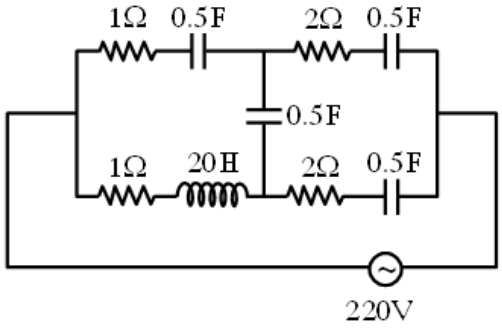
(1) 77 Ω, 97.6 Ω

(2) 32 Ω, 68 Ω

(3) 100 Ω, 77 Ω

(4) 77 Ω, 68 Ω

26) At very high frequencies, the effective impendance of the given circuit will be____Ω.



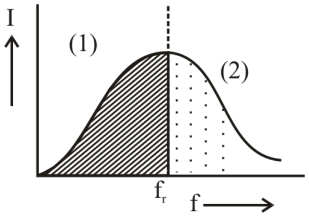

(1) 1

(2) 2

(3) 5

(4) 10

27) **Assertion (A) :-** For series RLC network, power factor of circuit in region (1) is positive and in region (2) is negative.
**Reason (R) :-** Overall nature of circuit in region (1) is inductive while in region (2) is capacitive.




(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

28) The r.m.s. current in an ac circuit is 2 A. If the wattless current be $\sqrt{3}$A, what is the power factor

(1) $\frac{1}{\sqrt{3}}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{2}$

(4) $\frac{1}{3}$

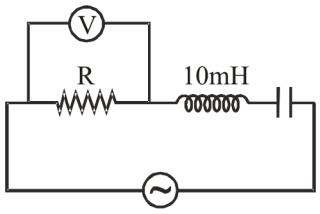
29) In a series A.C. circuit the current and voltage are given respectively as :
$I = 10 \sin(\omega t)$
$V = 10 \sin(\omega t + \pi/4)$
then which of the following element may be present in the circuit :-
(a) Only C (b) All L, C and R
(c) Both C and R (d) Both L and R

(1) a, c

(2) a, b, c

(3) b, d

(4) b, c




30) In the circuit of Figure, the volmeter reads 75 V.Value of C is :

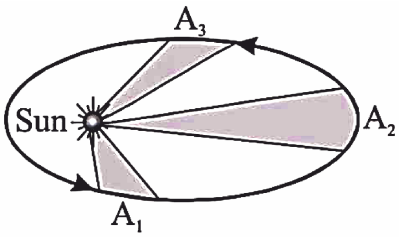
(1) 4 μF

(2) 2 μF

(3) 6 μF

(4) 10 μF

31) A planet moving around sun sweeps area $A_1$ in 2 days, $A_2$ in 3 days and $A_3$ in 6 days. Then the relation between $A_1$, $A_2$ and $A_3$ is :




(1) $3A_1 = 2A_2 = A_3$

(2) $2A_1 = 3A_2 = 6A_3$

(3) $3A_1 = 2A_2 = 6A_3$

(4) $6A_1 = 3A_2 = 2A_3$

32) A tunnel is dug inside the earth across one of its diameters. A particle is projected inside the tunnel with velocity $\sqrt{\frac{GM}{R}}$ from one of its ends. Then maximum velocity attained by the particle in the subsequent motion is (assuming tunnel to be frictionless)

(1) $\sqrt{\frac{GM}{R}}$

(2) $\sqrt{\frac{2GM}{R}}$

(3) $\sqrt{\frac{3}{2}\frac{GM}{R}}$

(4) $2\sqrt{\frac{GM}{R}}$

33) The gravitational field in a region is given by $\vec{E}$ = 5 N/kg$\hat{i}$ + 12 N/kg $\hat{j}$. If the origin is taken as zero potential energy, then the potential energy at (0, 5) for 1 kg mass is :-

(1) –60 J

(2) 60 J

(3) –30 J

(4) None of these

34) The height at which the acceleration due to gravity decreases by 36% of it's value on the surface of the earth is :- (Assume radius of earth is R)

(1) $\frac{R}{4}$

(2) $\frac{R}{6}$

(3) $\frac{R}{2}$

(4) 4R

35) The value of 'g' at a particular point on the surface of earth is 9.8 $m/s^2$. Suppose the earth suddenly shrinks uniformly to half its present size without losing any mass. The value of 'g' at the same point (assuming that the distance of the point from the centre of earth does not shrink) will now be :-

(1) 4.9 $m/sec^2$

(2) 39.2 $m/sec^2$

(3) 9.8 $m/sec^2$

(4) 19.6 $m/sec^2$

36) A satellite can be in a geostationary orbit around a planet at a distance r from the centre of the planet. If the angular velocity of the planet about its axis doubles, a satellite can now be in a geostationary orbit around the planet if its distance from the centre of the planet is:

(1) $\frac{r}{2}$

(2) $\frac{r}{2\sqrt{2}}$

(3) $\frac{r}{(4)^{1/3}}$

(4) $\frac{r}{(2)^{1/3}}$

37) Two planets assumed to have spherical shape have the same mass but densities in the ratio 1:8. For these planets, which of the following statement (s) is/are true ?
(a) Acceleration due to gravity at planets surface will be in the ratio of 1:4.
(b) Acceleration due to gravity at planets surface will be in the ratio of 4:1.
(c) Escape velocities from their surfaces will be in the ratio $\sqrt{2}$ : 1.
(d) Escape velocities from their surfaces will be in the ratio 1: $\sqrt{2}$.

(1) a and d

(2) d and c

(3) only a

(4) only c

38) Two satellites $S_1$ & $S_2$ revolve around a planet in the same direction in circular orbits. Their period of revolution are 1 hour & 8 hour respectively. The radius of $S_1$ is $10^4$ Km. The velocity of $S_2$ with respect to $S_1$ will be :-

(1) $\pi \times 10^4$ Km/hr

(2) $\frac{\pi}{3} \times 10^4$ Km/hr

(3) $2\pi \times 10^4$ Km/hr

(4) $\frac{\pi}{2} \times 10^4$ Km/hr

39) Escape velocity of a body of 1 kg mass on a planet is 100 m/sec. Gravitational Potential energy of the body at the Planet is -

(1) - 5000 J

(2) - 1000 J

(3) - 2400 J

(4) 5000 J

40) A research satellite of mass 200 kg circles the earth in an orbit of average radius $\frac{3R}{2}$, where R is the radius of the earth. Assuming the gravitational Force on a mass of 1 kg on the earth's surface to be 10 N, the gravitational force on the satellite will be (approximately) :

(1) 780 N

(2) 889 N

(3) 989 N

(4) 689 N

41) A satellite is put in a circular orbit of radius R. Another satellite is thrown in a circular orbit of radius 1.01r. The percentage difference between the periodic time of second satellite with respect to the first will be.

(1) 1% increased

(2) 1% decreased

(3) 1.5% increased

(4) 1.5% decreased

42) Two particles of combined mass M, placed in space with certain separation, are released. Interaction between the particles is only of gravitational nature and there is no external force present. Acceleration of one particle with respect to the other when separation between them is R, has a magnitude :

(1) $\frac{GM}{2R^2}$

(2) $\frac{GM}{R^2}$

(3) $\frac{2GM}{R^2}$

(4) Not possible to calculate due to lack of information

43) Two concentric shells have masses M and m and their raddi are R and r respectively where R > r, if x be the distance from the common centre, then what is the gravitational potential at a point for which r < x < R :

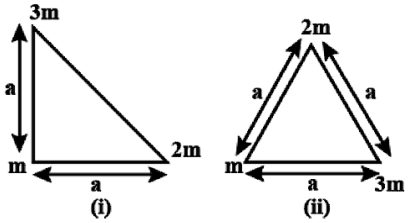
(1) $-G\left[\frac{M}{x}+\frac{m}{r}\right]$

(2) $-G\left[\frac{M}{x}-\frac{m}{r}\right]$

(3) $-G\left[\frac{M}{R}+\frac{m}{x}\right]$

(4) $-G\left[\frac{M}{R}-\frac{m}{x}\right]$

44) Consider two configurations of a system of three particles of masses m, 2m and 3m. The work done by gravity in changing the configuration of the system from figure (i) to figure (ii) is :




(1) Zero

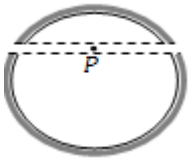
(2) $\frac{6Gm^2}{a}\left\{1+\frac{1}{\sqrt{2}}\right\}$

(3) $\frac{6Gm^2}{a}\left\{1-\frac{1}{\sqrt{2}}\right\}$

(4) $\frac{6Gm^2}{a}\left\{2-\frac{1}{\sqrt{2}}\right\}$

45)

A spherical shell is cut into two pieces along a chord as shown in the figure. *P* is a point on the plane of the chord. The gravitational field at *P* due to the upper part is $I_2$ and that due to the lower part is $I_1$ What is the relation between them




(1) $I_1 > I_2$

(2) $I_1 < I_2$

(3) $I_1 = I_2$

(4) No definite relation

ANSWER KEYS

# PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **A.** | **3** | **1** | **4** | **1** | **2** | **1** | **3** | **2** | **1** | **4** | **1** | **4** | **4** | **3** | **3** | **3** | **1** | **2** | **3** | **2** |
| **Q.** | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| **A.** | **1** | **3** | **1** | **3** | **1** | **2** | **4** | **3** | **3** | **4** | **1** | **2** | **1** | **1** | **3** | **3** | **1** | **1** | **1** | **2** |
| **Q.** | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| **A.** | **3** | **2** | **3** | **3** | **3** | | | | | | | | | | | | | | | |

SOLUTIONS

PHYSICS

1)

We know Q = CV
Hence $(Q_1)_{max}$ = 6 mC while $(Q_2)_{max}$ = 12mC
However in series charge is same so maximum charge on $C_2$ will also be 6mC (and not 12mC) and hence potential difference across $C_2$ will be
$V_2 = \frac{6mC}{3\mu F} = 2kV$ and as in series
$V = V_1 + V_2$
So $V_{max}$ = 6kV + 2kV = 8kV

2) Energy loss = $\frac{C_1C_2}{2(C_1+C_2)}(V_1 - V_2)^2$
$= \frac{10 \times 50 \times 10^{-12}}{2(10+50) \times 10^{-6}}(50-0)^2$
$= 1.04 \times 10^{-2}$ J

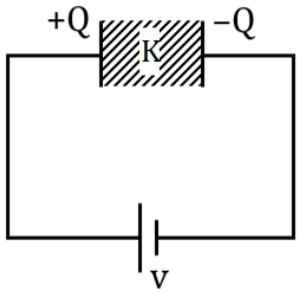
3)

Here $R_{eq} = R' = \frac{7R}{4}$
Now $i = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$
$\Rightarrow \frac{1}{4}\frac{q_0}{R'C} = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$
or $\ln 4 = \frac{t}{R'C}$
$\Rightarrow t = \frac{7RC}{4}\ln 4 = \frac{7RC}{2}\ln 2$

4) During the growth of voltage in a C-R circuit the voltage across a capacitor at time t is given by
$V = V_0\left(1 - e^{\frac{-t}{CR}}\right)$ for the given circuit as per given conduction at time t.
V = $\frac{3}{4}$ th of the voltage applied across C = $\frac{3}{4}V_0$
So, $\frac{3}{4}V_0 = V_0\left(1 - e^{-\frac{t}{RC}}\right)$
$\Rightarrow e^{-\frac{t}{RC}} = \frac{1}{4} \Rightarrow \frac{t}{RC} = \ln 4$
$\Rightarrow$ t = 2RC ln 2
$= 2 \times (2.5 \times 10^6) \times (4 \times 10^{-6}) \times (0.693) = 13.86$s

5)

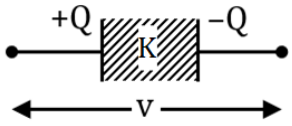
(1)




Q = (KC)V

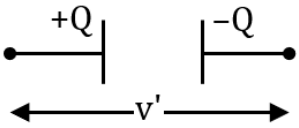
$\left\{\text{Where, } C = \frac{\varepsilon_0 A}{d}\right\}$

(2)




as battery is removed so charge will remain constant

(3)




After removing dielectric

Q = Q' = KCV

New voltage

$V' = \frac{KCV}{C} = KV$

{C → capacitance without dielectric}

6)

Initial energy of capacitor of capacitance 4 μF is

$$U_i = \frac{1}{2} \times (4 \times 10^{-6})(80)^2 = 0.0128 \text{ J}$$

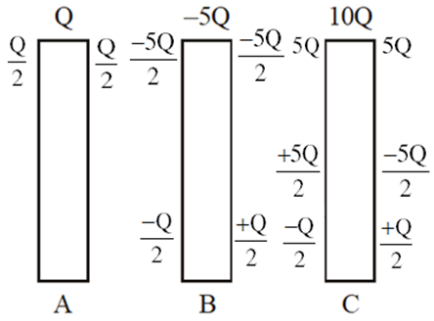
Final potential on 4μF capacitor after connection is

$$V = \frac{4 \times 80 + 6 \times 30}{4 + 6} = 50 \text{ V.}$$

So final energy on it

$$U_f = \frac{1}{2} \times 4 \times 10^{-6} (50)^2 = 0.005 \text{ J}$$

Energy lost by this capacitor = $U_i - U_f$ = 7.8 mJ




7)

$$(q_{in})_C = 5Q - \frac{Q}{2} + \frac{5Q}{2}$$

$$= \frac{14Q}{2} = 7Q$$

$$(q_{out})_C = 5Q + \frac{Q}{2} - \frac{5Q}{2}$$

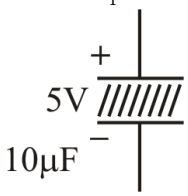
$$= +\,3Q$$

8) Power $= \dfrac{\text{Energy}}{\text{Time}} = \dfrac{\frac{1}{2}CV^2}{t}$ $= \dfrac{(40 \times 10^{-6})(3000)^2}{2 \times (2 \times 10^{-3})}$

$= 90$ kW

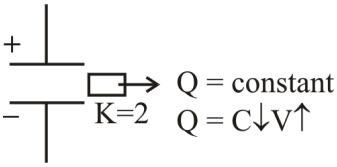
9)

When $K_1$ closed d, $K_2$ open




when $K_1$ open

K = 2 is removed and inserted in 4μF

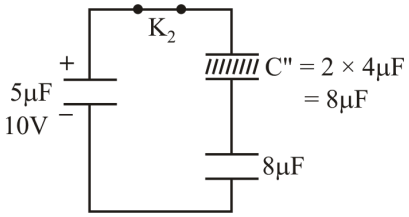



V' = KV

$= 2 \times 5 = 10$V

$$C' = \frac{10}{2} = 5\mu F$$

$K_2$ is closed

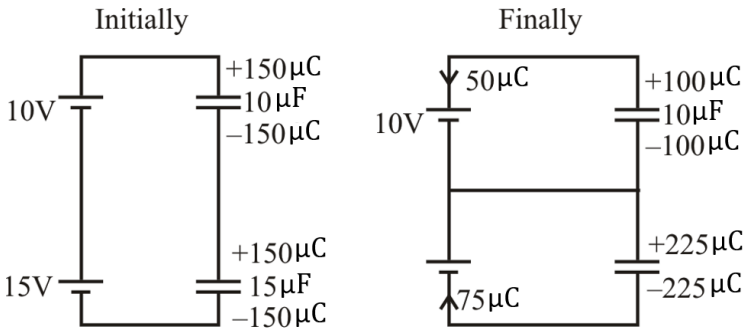



All are in series

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} + \frac{1}{5} = \frac{18}{40}$$

$$C_{eq} = \frac{40}{18}\mu F$$

Q = $\left(\frac{40}{18} \times 10\right) \mu C = \frac{200}{9} \mu C$

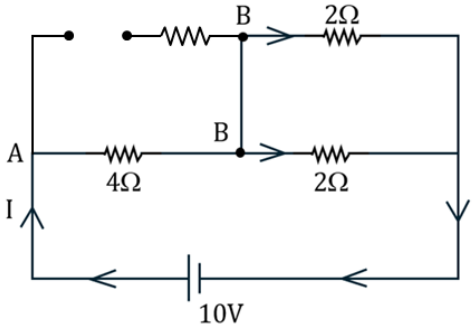



10)

Hence work done by one battery will be + and other will be – & since
(work by battery 1) + (work by battery) + ΔU ¹ 0

11)

In the steady state condition, no current will flow in the branch having capacitor therefore, the voltage across capacitor will be equal to the voltage across 4Ω Resistor.




The current passing through 4Ω resistor

$I = \frac{10}{4 + \frac{2}{2}} = \frac{10}{4+1} = \frac{10}{5} = 2A$

Therefore, the potential drop between points A and B

$V_{AB}$ = IR = (2) (4) = 8V

Therefore, the charge stored on capacitor

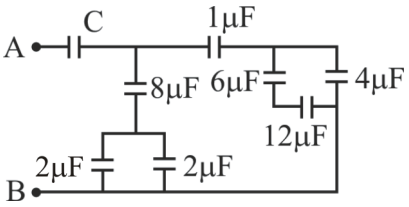
$q_{stored} = CV_{AB}$ = (3μF)(8) = 24 μC

12) $W = U_f - U_i = \frac{(CV_0)^2}{2\left(\frac{C}{3}\right)} - \frac{(CV_0)^2}{2C} = CV_0^2$

$= \frac{\epsilon_0 A V_0^2}{d}$ .

13)

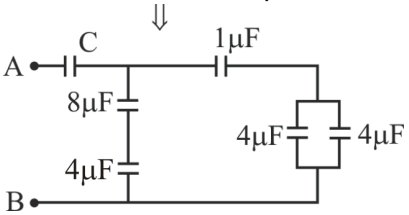
given $C_{AB}$ = 1μF




6μF and 12 μF in series

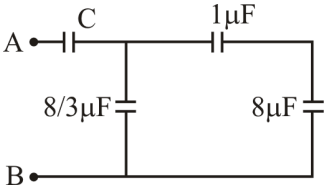
$$C_{eq} = \frac{6 \times 12}{6 + 12} = 4\ \mu F$$

$\Downarrow$



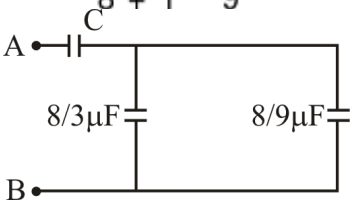

8μF and 4μF in series

$$C_{eq} = \frac{8 \times 4}{8 + 4} = \frac{8}{3}\mu F$$




8μF and 1μF in series

$$C_{eq} = \frac{8 \times 1}{8 + 1} = \frac{8}{9}$$

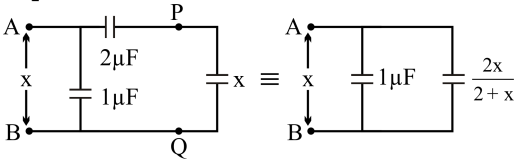



on further solving

$$C = \frac{32}{23}\mu F$$

14)

Equivalent CKt.




Let equivalent capacitance between A & B is x.

$$x = 1 + \frac{2x}{2 + x}$$

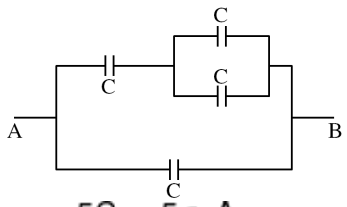
$$x^2 - x - 2 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4(1)(-2)}}{2}$$

x = 2μF

15)

Equivalent circuit




$$C_{eq} = \frac{5C}{3} = \frac{5\epsilon_0 A}{3d}$$

16)

Energy losses $\Delta U = \frac{1}{2} CV^{02} - \frac{1}{2} CV^2$

Fractional energy loss

$$\frac{\Delta U}{U_0} = \frac{\frac{1}{2}CV_0^2 - \frac{1}{2}CV^2}{\frac{1}{2}CV_0^2} = \frac{V_0^2 - V^2}{V_0^2}$$

$$= \frac{(100)^2 - (80)^2}{(100)^2} = \frac{20 \times 180}{(100)^2} = \frac{9}{25}$$

17) $Q_{1max} = 3\ C \times 10^3 C.$

$Q_{2max} = 4\ C \times 10^3 C.$

$Q_{max}$ for first branch $3C \times 10^3\ C$

$$V_{max_1} = \frac{3C \times 10^3 \times 5C}{6C^2} = \frac{5}{2} KV$$

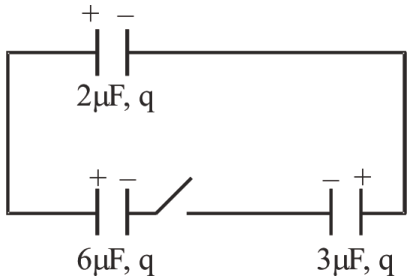
Similarly for second branch

$Q_{3_{max}} = 7C \times 10^3 C$ $\quad$ $Q_{4_{max}} = 6C \times 10^3 C$

$$V_{max_1} = \frac{6C \times 10^3}{21C^2} \times 10C = \frac{20}{7} KV$$

The two branches are in parallel. So in order to find max value of voltage for which no capacitor breaks down

$V_{max_1} < V_{max_2}.$




18)

Charge will flow by potential difference (P.d.).

$$\Delta q = \frac{2q}{3}$$

19) $Q = \frac{\omega}{\Delta\omega} = \frac{\omega L}{R} \Rightarrow \Delta\omega = \frac{50}{5} = 10$ rad/s

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5 \times 20 \times 10^{-6}}} = 100$ rad/s

$\omega_{min} = \omega^0 - \frac{\Delta\omega}{2} = 100 - \frac{10}{2} = 95$ rad/s

$\omega_{max} = \omega_0 + \frac{\Delta\omega}{2} = 100 + \frac{10}{2} = 105$ rad/s

20)

$\tan\phi = \frac{X_L}{R} = \frac{2\pi fL}{R}$

$= \frac{2 \times \frac{22}{7} \times 50 \times 0.7}{220} = \frac{220}{220} = 1$

$\phi = 45^\circ$

$i = \frac{V}{Z} = \frac{220}{200\sqrt{2}} = \frac{1}{\sqrt{2}}$ Amp.

Wattless current = $i\sin\phi$

$= \frac{1}{\sqrt{2}} \sin\frac{\pi}{4} = \frac{1}{2}$

21) **Explanation:**
To match the column-I statement to column- II statements.

**Concept:**
This question is based on Inductance, resistance, impedance of coil reactance of capacitor.

**Solution:**
Column - I
(A) Inductance of coil → depends on shape
→ depends on material inserted
so [A] → q, r

(B) Capacitance of capacitor → depends on shape
→ depends on material interested
so [B] → q, r

(C) impedance of coil → depends on resistivity
($X_L = 2\pi fL$) → depends on shape
→ depends on medium inserted
→ depends on external AC source so [C] → p, q, r, s
(D) Reactance of capacitor → depends on shape
→ depends on medium
→ depends o external AC voltage source $\left[X_C = \frac{1}{2\pi fc}\right]$
so [D] → q, r, s

**Final Answer:**
Option (1)

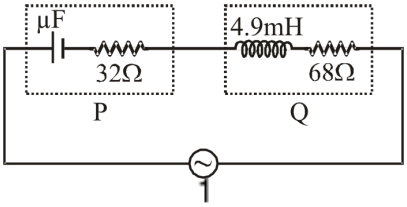
22) $I = 12 + 3 \sin \omega t + 4 \cos \omega t$
$I = 12 + 5 \sin (\omega t + \phi)$
Peak value = 17
Instantaneous value (t = 0)
$= 12 + 3 \times 0 + 4 \times 1 = 16$
Effective current = $\sqrt{12^2 + \frac{5^2}{2}} = \sqrt{\frac{313}{2}}$
Amplitude = 5

23)

In given question series LCR circuit is in resonance condition i.e $X_L = X_C$
Resonance angular frequency
$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 10^{-3} \times 20 \times 10^{-6}}} = 2500 \text{rad/sec}$
Impedance (z) = R(min) = 44Ω
Current $(I_{rms}) = \frac{V_{rms}}{Z} = \frac{220}{44} = 5\,A$
amplitude of current $(I_0) = 5\sqrt{2}\,A$

24) With dc : $P = \frac{V^2}{R} \Rightarrow R = \frac{(10)^2}{20} = 5\Omega;$
With ac : $P = \frac{V_{rms}^2 R}{Z^2} \Rightarrow Z^2 = \frac{(10)^2 \times 5}{10}$
Also, $Z^2 = R^2 + 4\pi^2 v^2 L^2$
$\Rightarrow 50 = (5)^2 + 4(3.14)^2 v^2 (10 \times 10^{-3})^2$
⇒ v = 80 Hz.

25) $\omega = \frac{1}{\sqrt{LC}}$




$\omega = \frac{1}{\sqrt{4.9 \times 10^{-3} \times 10^{-6}}} = \frac{1}{7 \times 10^{-5}}$
$X_L = \omega L = \frac{1}{7 \times 10^{-5}} \times 4.9 \times 10^{-3} = 0.7 \times 10^2 = 70\Omega$
$X_C = \frac{1}{\omega C} = 70\ \Omega$
$Z_P = \sqrt{R^2 + X_C^2} = \sqrt{32^2 + 70^2} \approx 77\ \Omega$
$Z_Q = \sqrt{R^2 + X_L^2} = \sqrt{68^2 + 70^2} \approx 97.6\ \Omega$

26)

$X_L = 2\pi fL$

f is very large

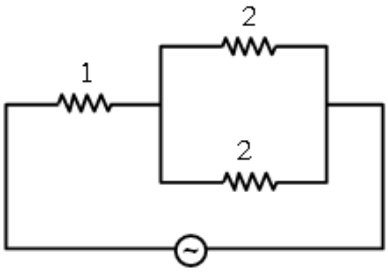
$\therefore X_L$ is very large hence open circuit.

$$X_C = \frac{1}{2\pi fC}$$

f is very large.

$\therefore X_C$ is very small, hence short circuit.

Final circuit




$$Z_{eq} = 1 + \frac{2 \times 2}{2+2} = 2$$

27)

In region (1) : $\omega < \omega_r \Rightarrow X_L < X_C \Rightarrow IX_C > IX_L$

$\Rightarrow V_C > V_L \rightarrow$ circuit with be capacitive (C–R)

In region (2) : $\omega > \omega_r \Rightarrow X_L > X_C \Rightarrow V_L > V_C$

$\rightarrow$ circuit with be capacitive (L–R)

The power factor :

$\text{Cos}\phi = \frac{R}{Z} \rightarrow$ always positive which is also true on the ohmic resistance, here power factor canot negative because the circuit cannot absorb power.

28)

$I_{rms} = 2A$

Wattless current = $\sqrt{3}A$

$\Rightarrow I_{rms} \sin\phi = \sqrt{3}$

$\Rightarrow 2\sin\phi = \sqrt{3}$

$\sin\phi = \frac{\sqrt{3}}{2} \Rightarrow \phi = 60^\circ$

Power factor = $\cos\phi$

$\cos 60^\circ = \frac{1}{2}$

29)

**Question Explanation :** For an AC circuit :

$I = 10 \sin(\omega t)$

$V = 10 \sin(\omega t + \pi/4)$

We need to identify the element(s) from the provided statements, which may be present in the circuit.

**Concept :** AC circuit
**Solution :** As we can see from the given equations that, the applied voltage (V) leads the current (I) by $\phi = \pi/4$ radians (or 45°).
The applied voltage leads the current by an angle $0° < \phi < 90°$ , only in -

A. an LR series AC circuit → Statement (d) is correct
B. an LCR series AC circuit, when inductance (L) dominates over capacitance (C) → Statement (b) is correct

Thus, statements (b) & (d) are correct.
**Final Answer :** Option (3)

30)
$$500 = \frac{1}{2\pi\sqrt{\left(10 \times 10^{-3}\right) C}}$$
$$\Rightarrow C = \frac{1}{10^{-2}\sqrt{\left(4\pi^2\right)\left(25 \times 10^4\right)}} = 10^{-5} = 10 \times 10^{-6}$$
F=10μF,
C = 10μF

31)

**Explain the Question :** Find relation b/w $A_1$, $A_2$ and $A_3$ as shown in figure.
**Concept :** This question is based on Kepler's 2nd law
**Solution :**
$$\frac{A}{t} = \text{constant}$$
$$\frac{A_1}{2} = \frac{A_2}{3} = \frac{A_3}{6}$$
Multiply with 6
$3A_1 = 2A_2 = A_3$
**Final answer :** 1

32) **Explain the question :** Find maximum velocity attained by body.
**Concept :** This question is based on conservation of mechanical energy.
**Solution :**
Maximum velocity will be at it's center
$K_A + U_A = K_C + U_C$
$$\frac{1}{2}m\left[\frac{GM}{R}\right] + m\left[\frac{-GM}{R}\right] = \frac{1}{2}mV_1^2 + m\left[-\frac{3}{2}\frac{GM}{R}\right]$$
$$\frac{1}{2}mV_1^2 = -\frac{GMm}{2R} + \frac{3}{2}\frac{GMm}{R}$$
$$V_1 = \sqrt{\frac{2GM}{R}}$$
**Final answer :** 2

33) $\Delta V = -\int \vec{E} \cdot d\vec{r}$

$= -\vec{E} \cdot \Delta\vec{r}$
$= -(5\hat{i} + 12\hat{j}) \cdot (0\hat{i} + 5\hat{j})$
= – 60 J/kg
$\Delta U = m\Delta V = 1(-60) = -60$ Joule

34) $g' = \frac{GM}{(R+h)^2}$
$= \frac{GM}{R^2\left(1+\frac{h}{R}\right)^2}$
$g' = \frac{GM}{R^2}\left[1+\frac{h}{R}\right]^{-2}$
$g' = g\left[1+\frac{h}{R}\right]^{-2}$
$\frac{64g}{100} = g\left(1+\frac{h}{R}\right)^{-2}$
$\frac{8}{10} = \left(1+\frac{h}{R}\right)^{-1}$
$1+\frac{h}{R} = \frac{10}{8}$
$\frac{h}{R} = \frac{2}{8} = \frac{1}{4}$
$h = \frac{R}{4}$

35) $g = \frac{GM}{R^2}$, Here M is mass of earth and R is distance of the given point from centre of earth. M and R both remain same so g remains same, 9.8 $m/s^2$.

36) $F = \frac{GMm}{r^2}$
$Mr\omega^2 = \frac{GMm}{r^2}$
$mr^3\omega^2 = GMm$
$r^3\omega^2 = \text{const}$
$r^3\omega^2 = (r')^3(\omega)^2$
$r^3\omega^2 = (r')^3(2\omega)^2$
$r^3\omega^2 = (r')^3(4\omega^2)$
$r' = \left(\frac{r^3}{4}\right)^{1/3}$
$r' = \frac{r}{(4)^{1/3}}$

37) $M_1 = M_2$
$\frac{4}{3}\pi R_1^3\rho_1 = \frac{4}{3}R_2^3\rho_2\pi$

$$\frac{\rho_1}{\rho_2} = \left(\frac{R_2}{R_1}\right)^3 = \frac{1}{8}$$

$$\frac{R_2}{R_1} = \frac{1}{2} \quad \text{....(1)}$$

(a) $g = \frac{GM}{R^2}$

$$\frac{g_1}{g_2} = \left(\frac{R_2}{R_1}\right)^2 = \frac{1}{4}$$

(d) $\nu = \sqrt{2gR}$

$$\frac{\nu_1}{\nu_2} = \left(\frac{g_1}{g_2}\right)^{1/2} \times \left(\frac{R_1}{R_2}\right)^{1/2}$$

$$= \left(\frac{1}{4}\right)^{1/2} \times \left(\frac{2}{1}\right)^{1/2}$$

$$= \frac{1}{\sqrt{2}}$$

38) $\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3} \Rightarrow \frac{1^2}{8^2} = \frac{(10^4)^3}{R_2^3}$

$\Rightarrow R_2^3 = 64 \times 10^{12}$

$\Rightarrow \boxed{R_2 = 4 \times 10^4 \text{ Km}}$

$V_{rel} = V_1 - V_2$

$$= \frac{2\pi R_1}{T_1} - \frac{2\pi R_2}{T_2}$$

$$= 2\pi\left(\frac{10^4}{1} - \frac{4 \times 10^4}{8}\right) = 2\pi \times \frac{10^4}{2}$$

$\Rightarrow \boxed{V_{rel} = \pi \times 10^4 \text{Km/hr}}$

39)

$E_{es} = -PE$

$E_{es} = \frac{1}{2}mv^2$

$E_{es} = \frac{-1}{2} \times 1(100)^2$

PE = –5000 J

40) **Question Explanation:**

Find the gravitational force acting on a 200kg satellite orbiting at a distance $\frac{3R}{2}$ from the center of the Earth, given that gravitational force on 1kg at Earth's surface is 10N.
**Concept:** This question is based on gravitational force on orbiting satellite.

**Solution:**

$$F = \frac{GM \times 1}{R^2} = 10\,N$$

$$F' = \frac{GM \times 200}{\left(\frac{3R}{2}\right)^2} = \frac{GM \times 200 \times 4}{9R^2}$$

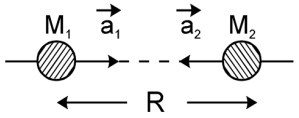
$$F' = \frac{10 \times 200 \times 4}{9} = 889\ N$$

**Final answer: Option (2)**

41) Correct option (3)

$T^2 \propto R^3$

42)




$$a_1 = \frac{GM_1M_2}{R^2/M_1} \qquad a_2 = \frac{GM_1M_2}{R^2/M_2}$$

$$\Rightarrow a_1 = \frac{GM_2}{R^2} \qquad a_2 = \frac{GM_1}{R^2}$$

$$a_{rel} = \vec{a}_1 - \vec{a}_2$$

$$= a_1 + a_2$$

$$= \frac{GM_2}{R^2} + \frac{GM_1}{R^2} = \frac{G(M_1 + M_2)}{R^2}$$

$$= \frac{GM}{R^2}$$

43) $$V = -\frac{GM}{R} + \left(-\frac{Gm}{x}\right) = -G\left[\frac{M}{R} + \frac{m}{x}\right]$$

44)

$$w = \Delta u = u_f - u_i$$

$$u_i = -\frac{2Gm^2}{a} - \frac{3Gm^2}{a} - \frac{6Gm^2}{\sqrt{2}a}$$

$$u_f = -\frac{3Gm^2}{a} - \frac{6Gm^2}{a} - \frac{2Gm^2}{a}$$

45) **Explain Question:**

Comparing gravitational fields inside a cut spherical shell.

**Concept:**

Gravitational field inside a shell.

**Solution:**

The net gravitational field at any point inside a complete uniform spherical shell is zero. When the shell is cut, the field from the upper segment ($I_2$) and the lower segment ($I_1$) must be equal

and opposite at any internal point P to result in a zero net field for the whole shell.

Calculation: $I_{net} = I_{upper} + I_{lower} = 0$
Therefore, magnitudes must be equal: $I_1 = I_2$.

**Final Answer:** Option (3)